

Feature	Open source product	Remarks
Public blockchain	Ethereum, Ripple, Litecoin, etc	Permissionless blockchain; participants are anonymous and proof of work as consensus algorithm; requires mining
Private blockchain	Hyperledger projects, Exonum, OpenChain, MultiChain	Permissioned blockchain; identity of the participants is known before; no need of mining and native cryptocurrency to pay for mining
Federated blockchain (consortium)	R3, EWF, etc	Group of companies or individuals coming together to make decisions for the best benefit of the whole network. Such groups are also called consortiums or a federated blockchain
Smart contracts in general programming languages	Hyperledger, Ethereum	Ethereum does not directly support Java; it does so with the help of a framework called web3j
No mining and native cryptocurrency	Hyperledger, Corda, Openchain	Typically, public blockchains require mining (as part of proof of work) to add a block to the chain. Miners will be rewarded using native cryptocurrency
Pluggable consensus algorithms	Hyperledger, Corda, OpenChain	Support for pluggable consensus used for different applications
Pluggable ordering service	Hyperledger	Support for pluggable ordering service used for different applications
Pluggable membership service	Hyperledger	Support for pluggable membership used for onboarding new users onto the network

- **Music industry:** Blockchain will make it easier to share music and use smart contracts to do so. The smart contracts will ensure that the revenue is distributed to the parties according to the agreement. It will also remove intermediaries.

Benefits of blockchain

- Greater transparency; transaction histories are becoming more transparent using blockchain technology
- Enhanced security; there are several ways that blockchain is more secure than other record-keeping systems
- Elimination of error handling through real-time tracking of transactions with no double spending
- Improved traceability
- Trusted record-keeping and shared trusted process
- Increased efficiency and speed, reduction of settlement time to mere seconds by removing intermediaries
- Reduced cost and complexity, material cost reduction through the elimination of expensive proprietary infrastructure
- Full automation of transactional processes, from payment through settlement
- Blockchain is not recommended for:
 - High performance (millisecond) transactions
 - Small organisations (no business network)
 - Database replacement
 - Messaging solutions

- Transaction processing replacement

Table 1 shows various open source blockchain products and their critical characteristics.

In summary, blockchain provides better security during transactions of any value. This unique and universal technology helps to streamline and automate nearly all customer services and legal contracts, while increasing the transparency and effectiveness of an enterprise. However, more work is needed for using blockchain technology to minimise enterprise costs, improve security in an era of cyber uncertainty and enhance customer delivery. **END** 

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